Weekly Progress Report - Chart Analysis AI

Report Date	2026-01-29
Author	That Le
Project	Chart Analysis AI V3
Current Phase	Phase 2 - Core Engine

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1. Executive Summary

Chart Analysis AI project now in phase 2, focusing on building the core engine for structured data extraction from chart images. The system employs a hybrid AI approach combining: - **Computer Vision** (YOLO) for chart detection and localization - **Geometric Analysis** (OpenCV + NumPy) for precise value extraction - **Small Language Model** (SLM) for OCR correction and reasoning

Key Achievements This Week:

Phase	Status	Complete percent
Phase 1: Foundation	DONE	100%
Phase 2: Core Engine	IN PROGRESS	~70%
Phase 3: Optimization	TODO	0%
Phase 4: Presentation	TODO	0%

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2. Data Collection Status

2.1. Academic Chart QA Dataset

Metric	Value
Total Chart Images	2,852
Total QA Pairs	13,297
Avg QA per Chart	4.66
Source PDFs	800+ papers (arXiv)
Generation Date	2026-01-24

2.2. Chart Type Distribution

Chart Type	Count	Percentage
Line	904	31.7%
Bar	598	21.0%
Scatter	409	14.3%
Other	271	9.5%

Chart Type	Count	Percentage
Heatmap	209	7.3%
Area	187	6.6%
Histogram	102	3.6%
Pie	96	3.4%
Box	76	2.7%

2.3. QA Type Distribution

Question Type	Count	Description
Structural	2,852 (21.4%)	Title, labels, legend, axes
Counting	2,833 (21.3%)	Counting elements
Comparison	2,558 (19.2%)	Highest, lowest, differences
Reasoning	2,534 (19.1%)	Trends, patterns, insights
Extraction	2,520 (19.0%)	Specific data values

2.4. Data Quality Notes

Hiện trạng: Đang tiến hành lọc lại dữ liệu do khối lượng lớn và cần đảm bảo chất lượng cho training: - Loại bỏ ảnh nhiễu (diagrams, photos, logos đã bị lọc nhầm) - Kiểm tra lại classification accuracy - Chuẩn hóa format QA pairs

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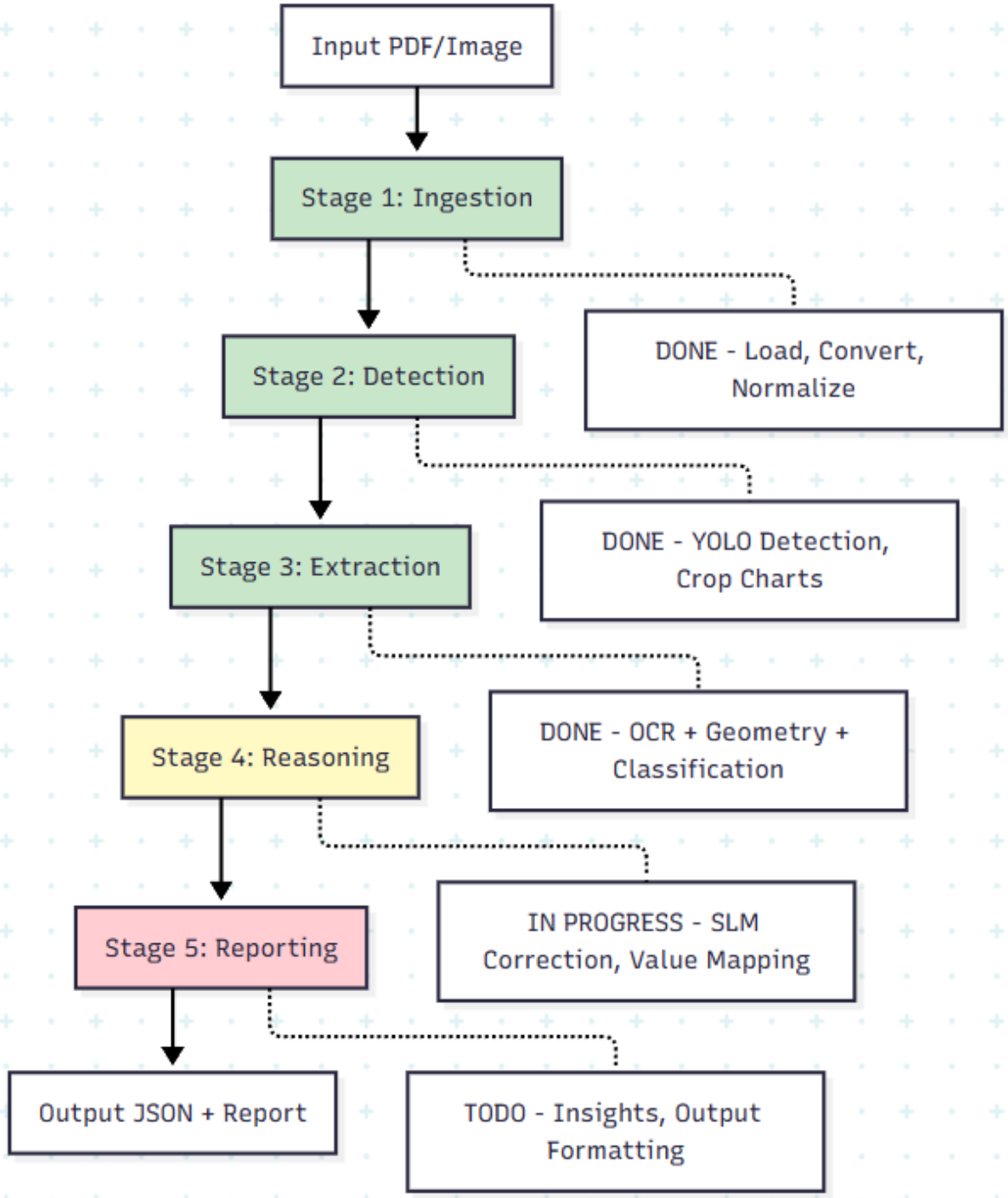
3. Pipeline Implementation Status

3.1. Overall Pipeline Architecture (5-Stage Model)



3.2. Stage-by-Stage Status

Stage 1: Ingestion DONE

Component	Status	Description
PDF Processor	Done	PyMuPDF (fitz) integration
Image Loader	Done	PNG/JPG support
Quality Validation	Done	Resolution, blur detection
Normalization	Done	Resize, contrast enhancement

File: s1_ingestion.py

Stage 2: Detection DONE

Component	Status	Description
YOLO Integration	Done	Ultralytics YOLOv8/v11
Confidence Filtering	Done	Threshold-based filtering
Multi-chart Handling	Done	Crop multiple charts per image
BBox Storage	Done	Save coordinates for traceability

File: s2_detection.py

Stage 3: Extraction DONE (Week 1 Focus)

Submodules implemented:

Submodule	File	Status	Description
Preprocessor	preprocessor.py	Done	Negative image + adaptive threshold
Skeletonizer	skeletonizer.py	Done	Lee algorithm, keypoint detection
Vectorizer	vectorizer.py	Done	RDP algorithm, subpixel refinement
OCR Engine	ocr_engine.py	Done	EasyOCR with role classification
Geometric Mapper	geometric_mapper.py	Done	Axis calibration, pixel-to-value
Element Detector	element_detector.py	Done	Bars, markers, pie slices
ResNet Classifier	resnet_classifier.py	Done	94.66% accuracy
ML Classifier	ml_classifier.py	Done	Production wrapper
Simple Classifier	simple_classifier.py	Done	Baseline fallback

Test Results (2026-01-29):

Metric	Value
Total Images Tested	800+ (100 per type)
Classification Accuracy	**100%** (all 8 types)
OCR Confidence	**91.5%** (EasyOCR)
Overall Confidence	**92.6%**
Avg Processing Time	~7.6s per image

ResNet-18 Classifier Highlights:

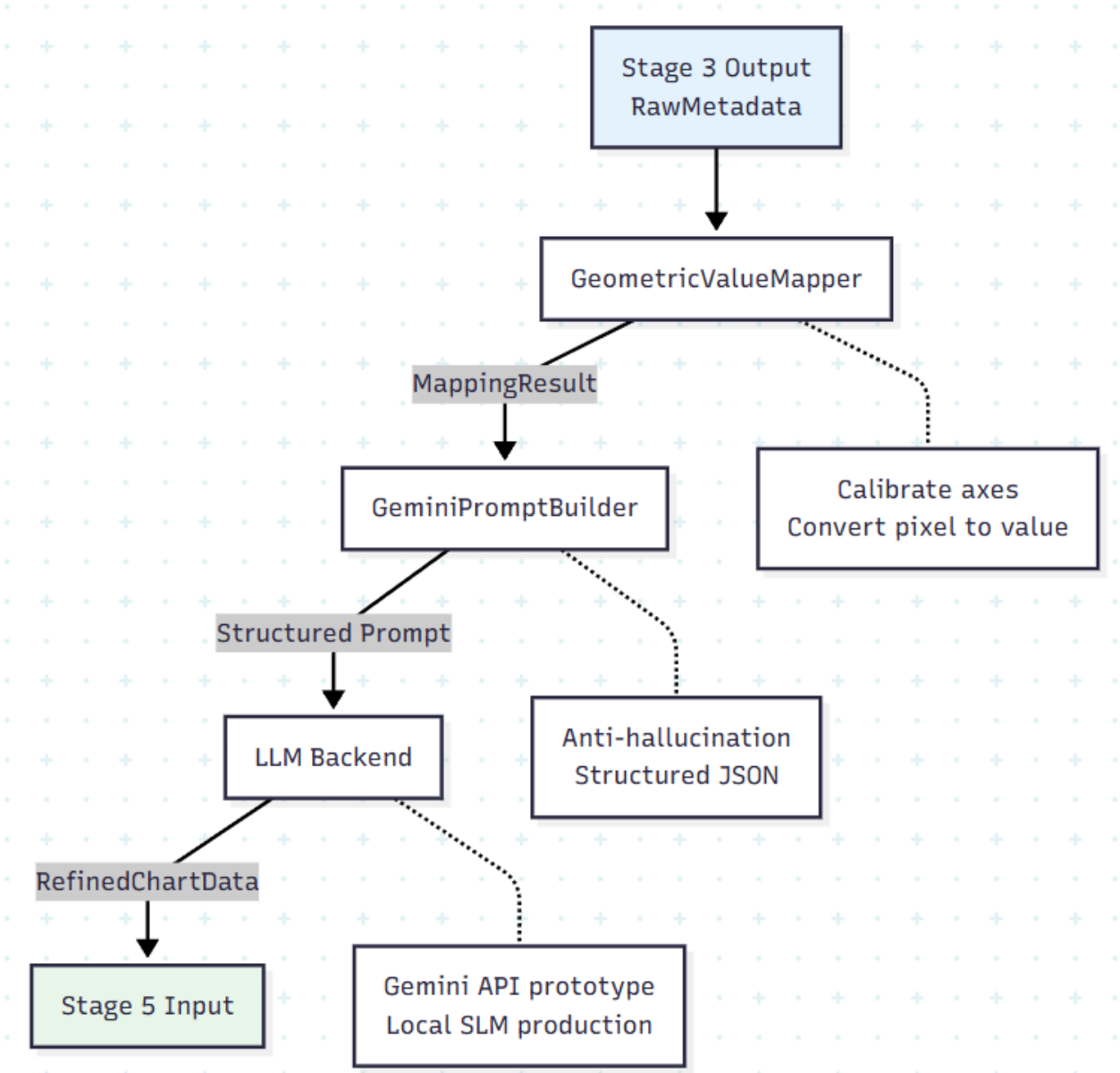
Metric	Value
Test Accuracy	94.66%
Integration Test	93.75% (15/16 samples)
Training Time	27 minutes (NVIDIA GPU)
ONNX Inference	6.90ms (CPU), 144.9 img/sec
Model Size	42.64 MB (ONNX)
Classes	8 types
Explainability	Grad-CAM visualizations

Stage 4: Reasoning IN PROGRESS (Current Week)

Implemented Components:

Component	File	Status	Description
GeometricValueMapper	value_mapper.py	Done	Pixel-to-value conversion
GeminiPromptBuilder	prompt_builder.py	Done	Canonical Format prompts
ReasoningEngine	reasoning_engine.py	Done	Orchestrator class
GeminiEngine	gemini_engine.py	Done	Gemini API integration
Prompt Templates	prompts/	Done	reasoning.txt, canonical_format.md
Unit Tests	test_s4_reasoning/	Done	36 test cases
Stage4Reasoning	s4_reasoning.py	Done	Main orchestrator
Local SLM	-	TODO	Qwen-2.5 / Llama-3.2

Architecture:



Gemini API Configuration: - Model: gemini-3.0-flash-preview - Temperature: 0.3 (deterministic) - Max tokens: 2048

Known Issues: - PaddleOCR 3.3.x has compatibility issues on Windows → Switched to EasyOCR - Gemini API occasionally returns unparseable JSON → Fallback mechanism works

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Stage 5: Reporting TODO

Planned Components:

Component	Description
Parallel Processing	ThreadPoolExecutor for batch processing

Component	Description
Insight Generation	Trend, comparison, anomaly, summary
JSON Validator	Schema validation (Pydantic)
Report Generator	Markdown + JSON output
Traceability Info	Source tracking

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4. Code Structure Overview

```
src/core_engine/
├── __init__.py
├── pipeline.py          # Main orchestrator
├── exceptions.py        # Custom exceptions
├── schemas/
│   ├── common.py        # BoundingBox, Color, SessionInfo
│   ├── enums.py         # ChartType enum
│   ├── extraction.py    # Extraction-specific schemas
│   ├── qa_schemas.py    # QA pair schemas
│   └── stage_outputs.py # Stage1-5 Output schemas
├── stages/
│   ├── base.py          # BaseStage abstract class
│   ├── s1_ingestion.py  # Stage 1 implementation
│   ├── s2_detection.py  # Stage 2 implementation
│   ├── s3_extraction/   # Stage 3 (12 files)
│   │   ├── s3_extraction.py
│   │   ├── preprocessor.py
│   │   ├── skeletonizer.py
│   │   ├── vectorizer.py
│   │   ├── ocr_engine.py
│   │   ├── geometric_mapper.py
│   │   ├── element_detector.py
│   │   ├── resnet_classifier.py
│   │   ├── ml_classifier.py
│   │   ├── simple_classifier.py
│   │   └── classifier.py
│   └── s4_reasoning/    # Stage 4 (7 files)
│       ├── s4_reasoning.py
│       ├── gemini_engine.py
│       ├── reasoning_engine.py
│       ├── prompt_builder.py
│       ├── value_mapper.py
│       └── prompts/
└── validators/          # Input validators
```

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5. Testing Status

Test Suite	Files	Cases	Status

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Schema Tests	test_schemas.py	~20	Pass
Stage 3 Tests	test_s3_extraction/	129	Pass
Stage 4 Tests	test_s4_reasoning/	36	Pass
Integration Tests	scripts/test_*.py	~10	Pass

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6. Next Steps (Định hướng)

Immediate (This Week)

1. **Complete Stage 4 Integration Testing**
    - Test full pipeline Stage 3 → Stage 4
    - Validate Gemini API output quality
    - Document edge cases
2. **Begin Stage 5 Implementation**
    - Design parallel processing architecture
    - Implement insight generation module
    - Setup JSON schema validation

Short-term (Next 2 Weeks)

1. **Local SLM Training**
    - Prepare training data from Gemini outputs
    - Fine-tune Qwen-2.5-1.5B or Llama-3.2-1B
    - Compare with Gemini baseline
2. **Performance Optimization**
    - Batch processing implementation
    - Caching mechanism for repeated queries
    - Memory optimization for large documents

Medium-term (Month 2)

1. **Benchmarking**
    - Compare with GPT-4V baseline
    - Evaluate on public chart datasets
    - Document accuracy metrics
2. **Demo & Documentation**
    - Streamlit demo interface
    - API documentation (FastAPI)
    - Thesis chapter drafts

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7. Technical Decisions Made

Decision	Rationale	Date

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ResNet-18 over SimpleClassifier	94.66% vs 37.5% accuracy	2026-01-25
EasyOCR over PaddleOCR	Windows compatibility	2026-01-26
Gemini API for prototyping	Fast iteration before local SLM	2026-01-26
Canonical Format prompts	Anti-hallucination, structured	2026-01-30
Stage 3 = Pure Geometry	No LLM dependency, reproducible	2026-01-29

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8. Risks & Mitigations

Risk	Impact	Mitigation
Gemini API rate limits	Slow development	Local SLM fallback
OCR accuracy on complex charts	Poor value extraction	Multi-engine fallback
Dataset bias (academic only)	Limited generalization	Plan diverse data collection
Training time for local SLM	Delay in production	Use distillation techniques

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9. Resources

Documentation

- MASTER_CONTEXT.md - Project overview
- PIPELINE_FLOW.md - Pipeline architecture
- STAGE3_EXTRACTION.md - Stage 3 details
- STAGE4_REASONING.md - Stage 4 details

Models

- ResNet-18 Weights: models/weights/resnet18_chart_classifier_best.pt
- ONNX Export: models/onnx/resnet18_chart_classifier.onnx
- Grad-CAM: models/explainability/gradcam_*.png

Notebooks

- 00_quick_start.ipynb - Quick demo
- 03_stage3_extraction.ipynb - Stage 3 exploration
- 04_stage4_reasoning.ipynb - Stage 4 demo

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10. Summary

What's DONE: - Phase 1 complete với 2,852 classified charts và 13,297 QA pairs - Stage 1-3 fully implemented và tested (100% classification accuracy) - ResNet-18 classifier với 94.66% accuracy - Stage 4 core components implemented (ValueMapper, PromptBuilder, GeminiEngine)

What's IN PROGRESS: - Stage 4 integration testing - Data quality refinement (lọc lại dataset) - Local SLM preparation

What's PLANNED: - Stage 5: Reporting implementation - Local SLM training (Qwen-2.5 / Llama-3.2) - Benchmarking và optimization

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Report created by That Le on 2026-01-29